

Check your exam details, then complete the fields below

Surname

Other names

Centre number

Candidate number

Advanced Level · Difficulty: Hard

Mathematics

Date 8 August 2026Duration 2h 30mPaper ref. EB-MATH-A-2026-014

You will need

Formula and statistical tables booklet, scientific or graphical calculator, ruler, pencil for diagrams.

125

Total marks

Before you start

- Write in **black pen**. Pencil is fine for graphs, diagrams and sketches — keep it dark enough to read (HB or B).
- Fill in your name, centre number and candidate number at the top of this page.
- Answer every question in the space given — *there's usually more room than you'll need*.
- Label multi-part answers clearly so the examiner can follow which part you're answering.

About this paper

- You'll find **18** questions on this paper, worth **125** marks in total.
- The mark allocation for each question is printed in brackets alongside it — use it to judge your pace.
- A calculator is permitted, but it must not perform symbolic algebra, calculus, or hold retrievable formulae.
- Give answers to three significant figures unless the question tells you otherwise.

Exam technique

- Read each question fully before you start writing — the last line often changes what's being asked.
- Show your working throughout. A correct answer with no method may not earn full marks.
- Attempt every question — a partial answer can still pick up method marks.
- Leave time at the end to check your answers, especially arithmetic and units.

1.

5

(4)

(b)

g

(4)

1
t

(1)

(2)

(5)

(a) Given $0 < a < 1$, sketch the curve with equation

3.

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y -axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

(b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

(c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

The population of a town was monitored.

4. Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

- (a) find the value of a and the value of b .

(3)

According to the model, exactly T years after monitoring began, the population was 85 000

Making your method clear,

- (b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)

Find

5.

$$\int \frac{4x^5 + 3}{2x^2} dx$$

giving your answer in simplest form.

(5)

1

(1)

(2)

(5)

上

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)

1	1
1	1

(a) Express $f(x)$ in the form

(3)

(2)

(1)

(1)

9. Given that $S_{\infty} = 3a$

(2)

$$u_2 - u_4 = 16$$

(b) find the value of S_{10} giving your answer to one decimal place.

(5)